Universitat Pompeu Fabra Barcelona
NeTS research group
Network Technologies and Strategies

Fairness in Collision-Free WLANs

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Motivation

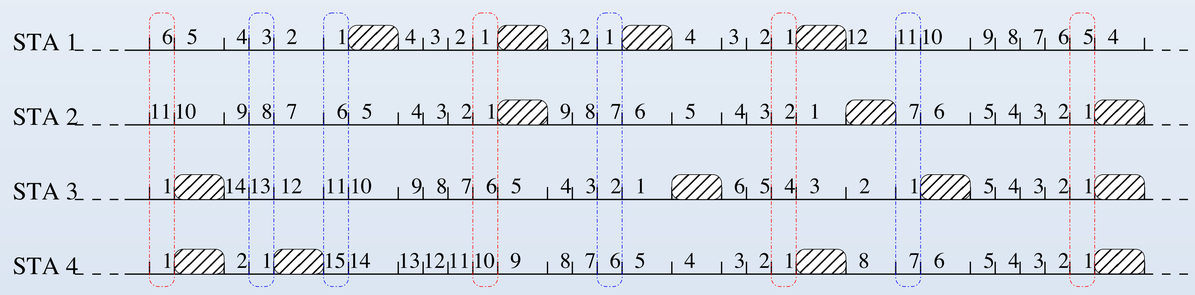
Wireless LANs are composed of nodes that must contend for the medium in a distributed manner. If two or more contenders attempt transmission at the same time, a collision occurs.
Collisions are a main cause of throughput degradation in Wireless Local Area Networks (WLANs), so by constructing collision-free WLANs one can attain greater levels of throughput.

1. CSMA/CA

Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) is the most widely used protocol for medium access control (MAC) in WLANs. CSMA/CA's job is to coordinate access to the medium for each contender.

When a node has something to transmit:

- Picks a random backoff counter $B \in [0, CW(k) - 1]$, where $CW(k) = 2^k CW_{\min}$ is the contention window, with $CW_{\min}$ its minimum value.
- Each passing empty slot decrements B by one. Contenders attempt transmission when the counter expires ($B = 0$).
- If there is a collision:
  - Colliding nodes increment the backoff stage (k) in one.
- After a successful transmission, the contender resets its backoff stage ($k = 0$).



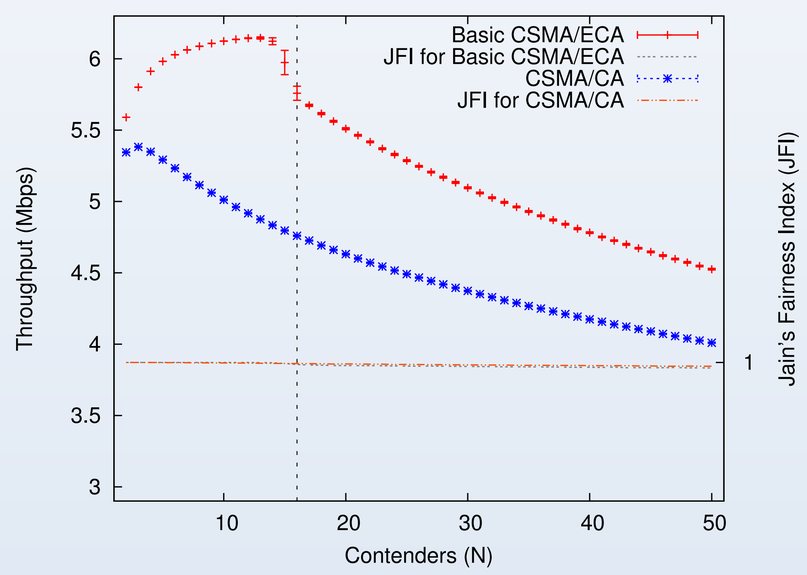
Example CSMA/CA behavior ($CW_{\min} = 16$).

2. Basic CSMA/ECA

CSMA/CA relies in a random backoff counter (B) which by its nature generates collisions. Furthermore, CSMA/CA instructs nodes to reset the backoff stage (k) after a successful transmission: increasing the collision probability.

Carrier Sense Multiple Access with Enhanced Collision Avoidance [1] (Basic CSMA/ECA):

- Picks a deterministic backoff counter $B_d = CW_{\min}/2$ after successful transmissions.
- Achieves a collision-free state.
- Basic CSMA/ECA's throughput goes beyond CSMA/CA's.



Throughput and fairness in CSMA/CA and Basic CSMA/ECA ($CW_{\min} = 32$).

Nevertheless, when the number of contenders surpasses $CW_{\min}/2$, the system incurs in a mixed behavior; some nodes pick a random and others a deterministic backoff counter. This setup has undesired repercussions in the attained throughput, approximating Basic CSMA/ECA's to CSMA/CA's.

3. CSMA/ECA + hysteresis and fair share

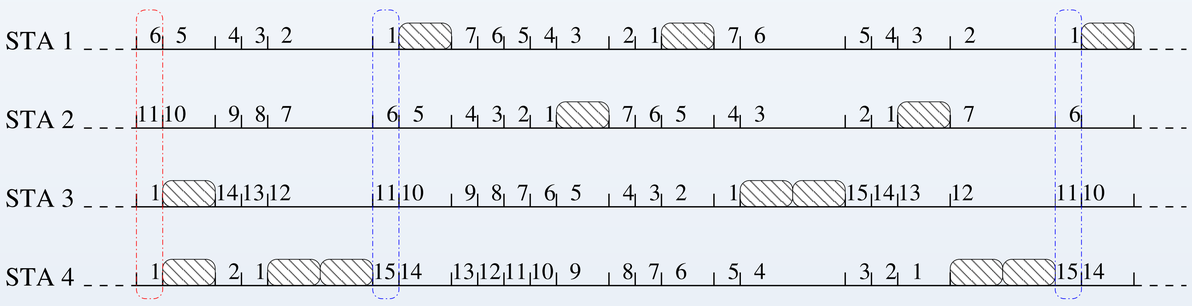
CSMA/ECA is totally distributed, that means that the number of nodes is unknown to all contenders. So, to make it possible to achieve a collision-free state with more than $CW\min/2$ contenders, CSMA/ECA:

- Instructs nodes **not** to reset their backoff stage after successful transmissions. It is reset only when the queue is empty.
- Picks a new deterministic backoff $B_d = CW(k)/2$.

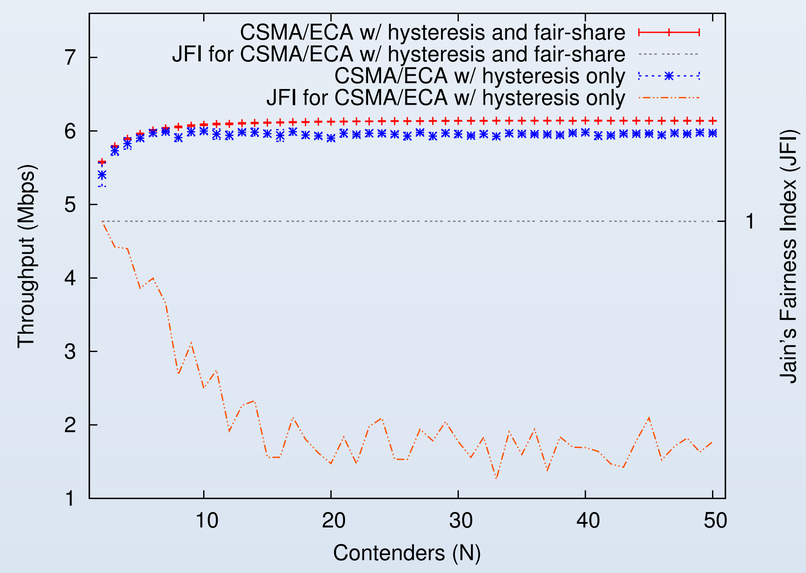
We called this measure *hysteresis*. With hysteresis some nodes may have larger B_d than others. This unfairness issue is averted by:

- Allowing nodes at backoff stage k to send 2^k packets.

This is called *fair share* [2].



Example CSMA/ECA behavior with hysteresis and fair share ($CW_{\min} = 16$).



Throughput and fairness when incorporating hysteresis and fair share to CSMA/ECA.

4. Conclusions, Future plans and Acknowledgements

Hysteresis allows CSMA/ECA to allocate any number of contenders in a collision-free state, while fair share compensates the unfairness issue; allowing CSMA/ECA to attain greater throughput than CSMA/CA under most typical conditions.
As future work, we plan to:

- Test CSMA/ECA under non-saturated scenarios.
- Implement IEEE 802.11e EDCA quality of service measures.
- Implement CSMA/ECA in cheap commodity hardware [3].

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References

[1] Barcelo, J. and Toledo, A.L. and Cano, C. and Oliver, M. Fairness and Convergence of CSMA with Enhanced Collision Avoidance (ECA). *2010 IEEE International Conference on Communications (ICC)* , may 2010, pp 1–6.

[2] Sanabria-Russo, L. and Barcelo, J. and Bellalta, B. Fairness in Collision-Free WLANs. *INFOCOM, 2013 Student Poster Abstract and ArXiv e-print* , February 2013.

[3] Tinnirello, I. and Bianchi, G. and Gallo, P. and Garlisi, D. and Giuliano, F. and Gringoli, F. Wireless MAC processors: Programming MAC protocols on commodity Hardware. *INFOCOM, 2012 Proceedings IEEE* , march 2012, pp 1269–1277.